Scratch Defect Segmentation via Spatial-Frequency Prompt

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Abstract

Scratch Defect Segmentation (SDS) aims to identify scratch defects amidst a product's inherent surface texture with mask prediction. Recent methods have achieved promising results by introducing prompt mechanisms for this task. However, these methods often rely solely on spatial-domain prompts, limiting their performance in handling the high morphological variability of scratches and their high visual similarity to background textures. To address this limitation, we propose **SFPSeg**, a novel Spatial-Frequency dual-domain prompt learning framework for robust scratch segmentation. To this end, a novel Mixed-domain Prompt Learning Decoder (MPLD) is introduced in SFPSeg to decode complementary and diverse patterns of scratches from both domains. Specifically, a novel Morphology-aware Spatial Prompting (MSP) module is introduced in MPLD to adaptively learn hierarchical morphological details from the spatial domain. Meanwhile, another novel module, namely Wavelet Prompt Filter (WPF), is introduced in MPLD to enhance scratch-related frequency components while suppressing background responses from the frequency domain. Comprehensive experiments on four SDS benchmarks demonstrate that SFPSeg outperforms state-of-the-art methods, achieving an mIoU of 91.16% on NEUSD, 81.40% on Severstal, 91.01% on Car, and 78.33% on Chip. The code is available at https://github.com/ShanWuJ/SFPSeg.

CCS Concepts

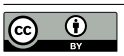
• **Computing methodologies** → **Visual inspection**.

Keywords

Scratch defect segmentation; Spatial-frequency prompt learning; Adaptive morphological modeling; Background responses suppression

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1 Introduction

Scratche Defect Segmentation (SDS) is a critical yet challenging quality control task in industrial manufacturing. It aims to precisely distinguish scratch regions amidst a product's inherent texture at the pixel level. Compared to bouding-box-level detections [17, 25], SDS provides fine-grained localization and shape information that is essential for severity assessment and sub sequent processes.

Most existing SDS methods, such as feature enhancement approaches [3, 7, 14], multi-task learning frameworks [33, 46], and global context guidance networks [43, 44, 49], primarily operate within image-label pairs. They focus on mining and refining visual patterns solely from the annotated defect data itself. Consequently, these models struggle to leverage richer, prior knowledge beyond the provided dataset. This inherent limitation hinders their robustness when encountering imaging variations, or defect manifestations that differ from the training set. Recent advances in prompt learning have shown promising results in image segmentation [20, 32, 36]. Building on this paradigm, subsequent works [9, 40, 51] have successfully introduced textual, interactive prompts (e.g., points, boxes), or adaptive task-specific learnable prompts for guided defect segmentation. Although effective, these prompt-based approaches rely solely on spatial-domain prompting, as illustrated in Fig. 1 (a), where prompt features are fused with visual features only at the spatial level. This spatially-confined prompting paradigm struggles to fully represent defects like scratches, which exhibit high morphological variability and strong visual similarity to background textures, thus revealing its performance bottleneck in such complex scenarios.

To overcome these limitations, we draw inspiration from prior works in frequency domain analysis [16, 23, 43, 53], which observe that objects exhibit distinctive and discriminative patterns in the frequency domain that can complement ambiguous spatial-domain features. Building on this insight, we propose **SFPSeg**, a novel Spatial-Frequency dual-domain Prompt learning framework for Scratch Defect Segmentation, as shown in Fig. 1 (b). Our framework is built upon a novel Mixed-domain Prompt Learning Decoder (MPLD), designed to capture complementary and diverse patterns of scratches from both domains. Specifically, MPLD consists of two core components: (1) A Morphology-aware Spatial Prompting

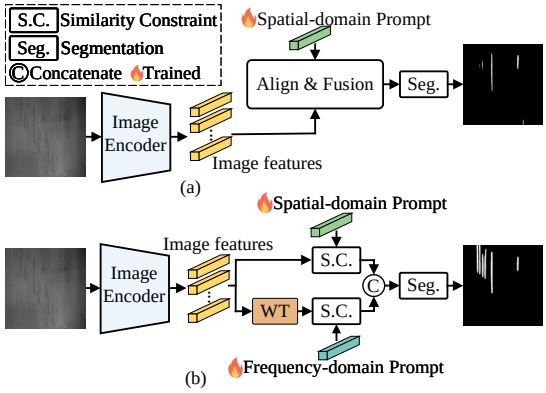


Figure 1: Comparisons between our method and previous methods. (a) Previous methods rely solely on spatial-domain prompts for segmentation. (b) Our method integrates complementary guidance from both spatial and frequency domains for more robust segmentation.

(MSP) module, which adaptively decodes scratch morphological details by fusing global Mamba-based features with gated multi-scale local features, followed by refinement via a prompt-based hybrid attention mechanism in the spatial domain. (2) A Wavelet Prompt Filter (WPF), which enhances scratch-related frequency components and suppresses background responses by introducing a learnable scratch-frequency prompt and measuring its similarity with frequency features. The outputs from MSP and WPF are then within MPLD in a parameter-free manner for precise mask prediction. Furthermore, we design a Bidirectional Feature Fusion (BFF) module to aggregate multi-level encoder features, thereby providing enriched multi-scale contextual information for MPLD.

Our contributions can be summarized as follows:

- We propose a novel SDS method. Unlike existing methods that primarily apply prompt learning in the spatial domain, SFPSeg introduces a novel spatial-frequency dual-domain prompt learning framework for SDS.
- To achieve this, we propose a novel MPLD that explicitly decodes scratch features from two complementary domains: the MSP module leverages a learnable prompt to adaptively aggregate and refine the morphological details of scratches; while the WPF employs a frequency-domain prompt to accentuate scratch-related components while attenuating irrelevant background responses.
- Comprehensive experiments conducted on four SDS datasets (NUESD, Severstal, Car, and Chip) demonstrate that SFPSeg significantly outperforms state-of-the-art (SOTA) SDS models, achieving mIoU scores of 91.16%, 81.40%, 91.01%, and 78.33%, respectively.

2 Related Work

2.1 Scratch Defect Segmentation

Existing SDS approaches can be broadly categorized into two main paths: traditional vision-based approaches and emerging prompt-based learning paradigms. Within traditional methods, researchers often rely on three key technical routes: feature enhancement [3, 7, 14, 47], multi-task learning frameworks [33, 46], and global context guidance mechanisms [43, 44, 49], all of which aim to mine features directly from image-label pairs. Nevertheless, their performance often suffers from poor generalization under imaging variations, primarily due to their inability to leverage external prior knowledge. Recently, prompt learning paradigms have shown significant potential in defect segmentation. Methods such as FAPrompt [51] and DefectSAM [40] have adapted textual or geometric prompts to guide the segmentation process. While effective at leveraging external knowledge, their sensitivity to formulation and reliance on manual prompt design hinder accurate, fully automated inspection. To mitigate these issues, RDPrompter [9] introduces adaptive, task-specific learnable prompts, moving the field toward greater automation. However, a fundamental limitation persists: these methods rely exclusively on spatial-domain prompting. Consequently, they struggle to capture the high morphological diversity of scratches and often fail to separate them from visually similar backgrounds. To overcome this, we propose a novel spatial-frequency dual-domain prompting framework to leverage complementary information from both domains for SDS.

2.2 Prompt Learning in Image Segmentation

Existing prompt-based segmentation methods can be categorized into three classes according to the type of prompt signal. (1) Text-based prompt methods: These approaches [21, 48, 51] typically freeze large-scale pre-trained models and use category names or descriptive language as guiding signals, achieving object recognition through the alignment and fusion of visual and textual semantic information. (2) Spatial prompt methods: These methods [31, 32, 40] integrate positional prompts—such as points, bounding boxes, clicks, or masks—with image features to segment target objects interactively. (3) Continuous prompt methods: These methods introduce learnable embedding vectors, such as separate learnable prompts [20], part-specific learnable prompts [15], or learnable universal prompts [45], as task-specific guidance.

2.3 Frequency Domain Analysis

Frequency domain analysis, typically relying on spectral transforms such as the Fourier transform (FT) [5], discrete cosine transform (DCT) [22], and wavelet transform (WT) [10], is widely used in computer vision tasks. FSEL [37] utilizes the global nature of the Fast Fourier transform to improve the discrimination of camouflaged objects. Compared to FT, WT can decompose features into different frequency components while preserving local spatial information. Methodologies such as WTConv [10] and BLENet [29] employ WT to expand the receptive field or enhance feature fusion within convolutional frameworks. Others, including MobileMamba [16], MFR [41], and PEFC-Net [53] leverage WT specifically to bolster the extraction of fine-grained, high-frequency features for superior local modeling. WPFormer [43] introduces wavelet-enhanced cross-attention to integrate frequency information for defect inspection.

3 Method

3.1 Preliminary

As shown in Fig. 2, the proposed SFPSeg adopts a backbone to extract four-level features $\{f_i\}_{i=1}^4 \in \mathbb{R}^{C\times H_i\times W_i}$ with 1/4, 1/8, 1/16, and 1/32 resolutions. Their channels are compressed to 64 via the CGS (a sequential *Conv-GN-SiLU* block). These features are then fused by a BFF module, yielding multi-scale fused features $\{X_i\}_{i=1}^4$. Finally, each X_i is fused with the upsampled output from the lower level and then fed into the MPLD for decoding.

3.2 Bidirectional Feature Fusion

The bidirectional feature fusion module refines features through complementary pathways: top-down semantic guidance and bottom-up detail enhancement. The top-down pathway starts from the highest-level feature f_4 and iteratively generates semantically enriched features $\{d_i\}_{i=1}^4$. Specifically, with d_5 initialized as 0, for levels $i = 4, 3, 2$, d_i is computed as $\mathrm{Up}_{2\times}(\mathrm{DWConv}(f_i + d_{i+1}))$. For level $i = 1$, d_1 is set to $f_1 + d_2$. $\mathrm{DWConv}(\cdot)$ denotes a depthwise convolution layer, and $\mathrm{Up}_{2\times}(\cdot)$ performs 2× upsampling. Subsequently, the bottom-up pathway aggregates spatial details to construct the feature pyramid $\{X_i\}_{i=1}^4 \in \mathbb{R}^{C\times H_i\times W_i}$ for decoding. It is initialized with $X_1 = d_1$ and iterates for $i = \{1, 2, 3\}$ as:

$$X_{i+1} = \mathrm{CGS}(\mathrm{concat}(\mathrm{DWConv}_{\downarrow}(X_i), d_{i+1})). \tag{1}$$

3.3 Morphology-aware Spatial Prompting Module

To address the highly variable morphology of scratches, we design the Morphology-aware Spatial Prompting (MSP) module in MPLD, as illustrated in Fig. 2 (b). The MSP integrates a global morphology perception path in parallel with a gated multi-scale local perception path. Their fused outputs are then refined by a Prompt-based Hybrid Attention (PHA) block, which performs adaptive cross-dimension feature modulation to achieve accurate segmentation. The input to the MSP is denoted as $F \in \mathbb{R}^{C\times H\times W}$.

The global morphology perception path is built upon the Mamba architecture [12], leveraging its efficient long-range modeling capability. The input F is first transformed into F_{norm} via a DWConv layer with a residual connection and Layer Normalization. Critically, to overcome the limitation of fixed scanning patterns for irregular scratches, we employ a 2D Dynamic Adaptive Scanning (DAS) method [24] prior to the State Space Model (SSM). DAS dynamically navigates F_{norm} based on local context, generating a sequential trace that follows scratch contours. This adaptive sequence is then processed by the SSM to capture its global morphological dependencies. Finally, the SSM output is reshaped, added to F_{norm} as a residual, and passed through BN and GELU to produce the global feature $f_g \in \mathbb{R}^{C\times H\times W}$. The process is formulated as follows:

$$F_{norm} = \mathrm{LN}(\mathrm{DWConv}(F) + F) \tag{2}$$

$$f_g = \mathrm{GELU}(\mathrm{BN}(\mathrm{SSM}(\mathrm{DAS}(F_{norm})) + F_{norm})) \tag{3}$$

The gated multi-scale local perception path employs two separate gating structures to process features at small and large scales separately. Each gating structure first computes a modulated feature h_k and a gating weight g_k in parallel, both in $\mathbb{R}^{C\times H\times W}$:

$$h_k = \mathrm{BottConv}_k^2(\mathrm{BottConv}_k^1(F)), \quad g_k = \mathrm{BottConv}_k^3(F) \tag{4}$$

where $\mathrm{BottConv}_k$ comprises two pointwise convolutions for channel adjustment, a DWConv with kernel size k, followed by Group Normalization and a SiLU activation function, and $k \in \{3, 5\}$. h_k is then dynamically calibrated via element-wise multiplication with g_k to selectively enhance informative regions and suppress redundant ones. This produces the gated feature flow G_k:

$$G_k = h_k \otimes g_k \tag{5}$$

where $\otimes$ denotes element-wise multiplication. This operation dynamically selects important feature flows, yielding the small-scale feature flow G_3 and the large-scale feature flow G_5. Finally, the two scale gated feature flows G_3 and G_5 are aggregated with the input feature F via a residual connection, producing the enhanced multi-scale local feature $f_l \in \mathbb{R}^{C\times H\times W}$:

$$f_l = G_3 + G_5 + F \tag{6}$$

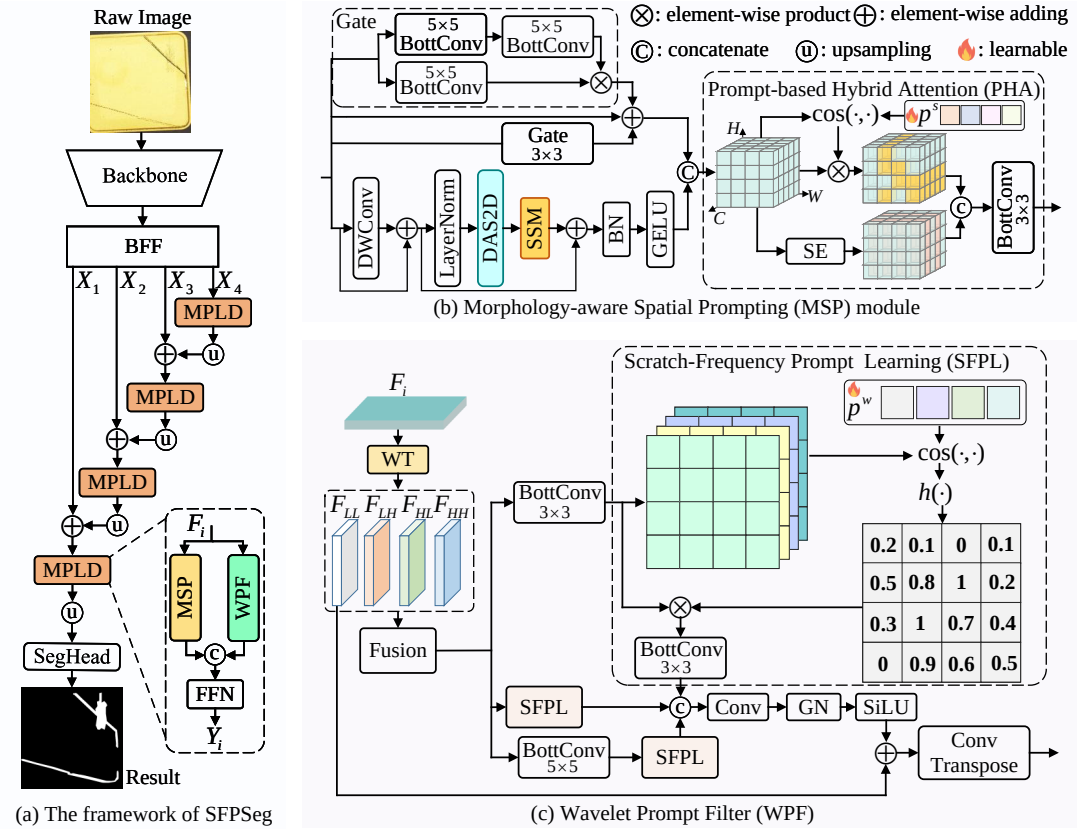
A Prompt-based Hybrid Attention (PHA) block then processes the concatenated global and local feature f_{cat} in two dimensions: channel and space. In the channel dimension, f_{cat} is processed by three parallel branches: an identity branch and two BottConv branches with kernel sizes 3 and 5, yielding channel multi-scale features $\{f_k | k \in \{cat, 3, 5\}\}$. Each feature f_k is independently enhanced by a Squeeze-and-Excitation (SE) block [18] to obtain its channel-attentive counterpart $\hat{f}_k$. These attentive features are aggregated with the original f_{cat} via residual summation to obtain the channel-modulated feature $\hat{f}_{cms}$. $\hat{f}_{cms}$ is further refined by a DWConv to produce the channel-refined feature $f_{cms} \in \mathbb{R}^{2C\times H\times W}$. In the spatial dimension, an analogous process constructs spatial multi-scale features $\{\tilde{f}_k | k \in \{cat, 3, 5\}\}$, where $f_{cat} = \tilde{f}_{cat}$. Each feature $\tilde{f}_k$ is modulated by a learnable scratch-specific prompt $P_k^s \in \mathbb{R}^{2C}$. Specifically, a dynamic spatial attention map $M_k^s \in \mathbb{R}^{H\times W}$ is computed by measuring the cosine similarity between $\tilde{f}_k$ and the prompt P_k^s along the channel dimension at each spatial location:

$$M_k^s = \frac{\langle \tilde{f}_k, P_k^s \rangle}{\|\tilde{f}_k\|_2 \|P_k^s\|_2} \tag{7}$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product, and $\|\cdot\|_2$ the Euclidean norm. Each attention map M_k^s is clamped to $[0, 1]$ and then applied to its corresponding feature $\tilde{f}_k$ via element-wise multiplication, yielding a spatially attentive feature $\hat{f}_k$ for each $k \in \{cat, 3, 5\}$. These features are concatenated and fused by a CGS block to produce the spatially modulated output $f_{sms} \in \mathbb{R}^{2C\times H\times W}$. Finally, the concatenation of f_{cms} and f_{sms} is processed by a BottConv for channel reduction and fusion, yielding the MSP output $F_{msp} \in \mathbb{R}^{C\times H\times W}$.

3.4 Wavelet Prompt Filter

Previous studies [37, 50, 53] have demonstrated that frequency-domain analysis can effectively discriminate texture that are difficult to distinguish in the spatial domain. Motivated by this insight, we propose a Wavelet Prompt Filter (WPF) in MPLD, as shown in Fig. 2 (c), to suppress background interference while enhancing scratch patterns in the frequency domain. Given its ability to decompose features into distinct frequency components while preserving local spatial information [10, 52], WT is adopted for frequency domain analysis in this work.



(a) The framework of SFPSeg

(b) Morphology-aware Spatial Prompting (MSP) module

(c) Wavelet Prompt Filter (WPF)

Figure 2: Overview of the proposed SFPSeg. Given an input image, multi-layer features are extracted from the backbone and fed into the Bidirectional Feature Fusion (BFF) module to produce fused multi-scale features $\{X_i\}_{i=1}^4$. X_i is then progressively decoded by the Mixed-domain Prompt Learning Decoder (MPLD). The decoder consists two key components: MSP (as detailed in (b)) and WPF (as detailed in (c)).

Given an input feature map $F \in \mathbb{R}^{C\times H\times W}$, it is decomposed via the 2D Haar wavelet transform into four frequency subbands $\{F_{LL}, F_{LH}, F_{HL}, F_{HH}\}$, each sized $\mathbb{R}^{C\times \frac{H}{2}\times \frac{W}{2}}$. An adaptive cross-band fusion module is then applied. The four subbands are concatenated and passed through a convolutional layer to generate adaptive weight maps $\{W_{LL}, W_{LH}, W_{HL}, W_{HH}\}$. Each subband is modulated by its corresponding weight via channel-wise multiplication, and the weighted subbands are summed to produce the fused frequency-domain feature $Z \in \mathbb{R}^{C\times \frac{H}{2}\times \frac{W}{2}}$:

$$Z = W_{LL} \otimes F_{LL} + W_{LH} \otimes F_{LH} + W_{HL} \otimes F_{HL} + W_{HH} \otimes F_{HH} \quad (8)$$

Z is processed by separate BottConv_3 and BottConv_5 layers, yielding local-context feature Z_{lc} and non-local-context feature Z_{nl} in $\mathbb{R}^{C\times \frac{H}{2}\times \frac{W}{2}}$. Z, Z_{lc}, and Z_{nl}, are then processed by three parallel Scratch-Frequency Prompt Learning (SFPL) blocks. Each SFPL block learns to adaptively enhance scratch-related frequency components while suppressing irrelevant background patterns.

The core of SFPL block corresponds to a learnable scratch-frequency prompt. Unlike textual or spatial prompts, ours is defined as a task-specific, randomly initialized token vector $P^w \in \mathbb{R}^C$ that operates purely in the parameter space. Given an input frequency feature (e.g., Z), we first reshape it to $\acute{Z} \in \mathbb{R}^{\frac{H}{2}\times \frac{W}{2}\times C}$, where each spatial location (h, w) corresponds to a C-dimensional feature vector. Next, the cosine similarity score between each vector in $\acute{Z}_{hw}$ and the scratch-frequency prompt P^w is computed via:

$$S_{hw} = \frac{\langle \acute{Z}_{hw}, P^w \rangle}{\|\acute{Z}_{hw}\|_2 \|P^w\|_2} \quad (9)$$

where $S_{hw} \in [-1, 1]$ denotes the similarity score at position (h, w). This yields a full spatial score map $S \in \mathbb{R}^{\frac{H}{2}\times \frac{W}{2}}$. Subsequently, S is normalized to the range $[0, 1]$ to obtain a soft attention mask M^w, which highlights regions whose frequency characteristics align with the learned scratch-frequency prompt. The mask M^w is then broadcast along the channel dimension and applied to the input feature Z via element-wise multiplication:

$$\tilde{Z} = Z \otimes M^w \quad (10)$$

This operation enhances positions with high similarity while suppressing those with low similarity, thereby amplifying scratch-related frequency patterns and attenuating uncorrelated background responses in the wavelet domain. Finally, a BottConv is applied to $\tilde{Z}$ for feature refinement and channel interaction, producing the the SFPL's output $\hat{Z} \in \mathbb{R}^{C\times \frac{H}{2}\times \frac{W}{2}}$. Correspondingly, for inputs Z_{lc} and

Z_{nl}, the outputs are $\hat{Z}_{lc}$ and $\hat{Z}_{nl}$. Subsequently, the outputs from the three SFPLs are concatenated along the channel dimension:

$$\hat{Z}_{\text{cat}} = \text{concat}(\hat{Z}, \hat{Z}_{lc}, \hat{Z}_{nl}) \tag{11}$$

yielding $\hat{Z}_{\text{cat}} \in \mathbb{R}^{3C\times\frac{H}{2}\times\frac{W}{2}}$. To fuse these features and reduce dimensionality, $\hat{Z}_{\text{cat}}$ is then processed by a CGS, producing the compact multi-scale representation $\hat{Z}_{\text{ms}} \in \mathbb{R}^{C\times\frac{H}{2}\times\frac{W}{2}}$. However, frequency-domain processing may attenuate certain structural details. Therefore, to compensate for this and enhance feature robustness, $\hat{Z_{ms}}$ is fused with the original low-frequency component F_{LL} via element-wise addition:

$$Z^{'}_{ms} = \hat{Z_{ms}} + F_{LL} \tag{12}$$

Finally, $Z^{'}_{ms}$ is upsampled back to the original spatial resolution $H \times W$ using a transposed convolution layer, generating the WPF's output $Z_{ms} \in \mathbb{R}^{C\times H\times W}$.

3.5 Decoding and Segmentation

Decoding proceeds from the deepest level (4th) to the shallowest, initialized with $Y_5 = \mathbf{0}$. At each level $i \in \{4, 3, 2, 1\}$, the input F_i of the MPLD is formed by fusing the skip connection feature X_i from the BFF with the upsampled output from the lower level Y_{i+1}:

$$F_i = X_i + \text{Up}_{2\times}(Y_{i+1}) \tag{13}$$

The decoder's output Y_i at this level is then computed as:

$$Y_i = \text{FFN}(\text{concat}(\text{MSP}(F_i), \text{WPF}(F_i))) \tag{14}$$

where FFN denotes a feed-forward network. Finally, the decoder's top-level output $Y_1 \in \mathbb{R}^{C\times\frac{H}{4}\times\frac{W}{4}}$ is fed into a segmentation head for mask prediction. The head first upsamples Y_1 by 2× and refines it with a SE block, producing $Y_{se} \in \mathbb{R}^{C\times\frac{H}{2}\times\frac{W}{2}}$:

$$Y_{se} = \text{SE}(\text{Up}_{2\times}(Y_1)) \tag{15}$$

This feature Y_{se} then undergoes a BottConv followed by another 2× upsampling, obtaining $out' \in \mathbb{R}^{C\times H\times W}$:

$$out' = \text{Up}_{2\times}(\text{BottConv}(Y_{se})) \tag{16}$$

Finally, a four-layer MLP F_{mlp} and a convolutional layer W are applied to out' to generate the final scratch segmentation prediction $S_{\text{pre}} \in \mathbb{R}^{1\times H\times W}$:

$$S_{\text{pre}} = \sigma(WF_{\text{mlp}}(out')) \tag{17}$$

where $\sigma(\cdot)$ denotes the sigmoid activation function.

3.6 Object Function

Following the previous work [26], we adopt a weighted combination of binary cross-entropy loss (L_{BCE}) and Dice loss (L_{Dice}) as our loss function. The total loss is formulated as follows:

$$L_{\text{total}} = \alpha L_{\text{BCE}}(S_{\text{pre}}, G) + (1-\alpha)L_{\text{Dice}}(S_{\text{pre}}, G) \tag{18}$$

where G denotes the ground truth and $\alpha \in [0, 1]$ is a hyperparameter that balances the contributions of the two losses. Empirically, we set $\alpha = 0.7$ for our experiments.

4 Experiments

4.1 Datasets and Evaluation Metrics

Datasets. To validate the proposed method, we conduct experiments on four scratch defect datasets: our in-house Chip-Scratch (**Chip**), and three public benchmarks (**NEUSD** [35], **Severstal** [11], and **Car** Paint [1]). Only scratch subsets are used for the latter three. Chip: 2741 expert-annotated samples of chip packaging surfaces (2192 train/549 test). NEUSD: 300 annotated images, split into 200 for training (augmented to 800) and 100 for testing. Severstal: 5150 scratch-defect images (4120 train/1030 test). Car: 335 scratch images (268 train/67 test). Chip and Severstal are particularly challenging due to the high visual similarity between their complex backgrounds and actual scratches.

Evaluation Metrics. Following prevalent practices in semantic segmentation [34, 38, 49], we adopt several widely-used metrics for evaluation: **Precision (P)** measures the proportion of predicted positives among all pixels predicted as positive. **Recall (R)** measures the proportion of actual positive pixels that are successfully retrieved. **F1** score is the harmonic mean of Precision and Recall. The mean Intersection over Union (**mIoU**) is the average of the Intersection over Union scores across all classes, serving as the key overall metric for segmentation performance.

4.2 Implementation Details

All experiments are conducted with PyTorch on a Tesla V100 GPU. We employ DAMamba-T [24], pretrained on ImageNet, as the backbone network. The proposed SFPSeg is trained for 150 epochs with a a batch size of 2, using the Adam optimizer. The learning rate is set to $1e-4$ for the backbone and $1e-3$ for other parts, and is decayed by a poly learning rate policy. A weight decay of 0.01 is applied. The random seed is fixed to 3407. Input images are resized to 512×512 for both training and inference.

4.3 Comparison with SOTAs on Different SDS datasets

We compare our SFPSeg with 18 SOTA methods, primarily including GCRANet [44], DCON [39], RodeconNet [42], TextSAM-EUS [36], DKD [46], SDCT [8], SRPCNet [3], WPformer [43], DBCNet [49], PEM [2], CamoDiffusion [4], FSEL [37], ZoomNext [30], IdeNet [13], MSCAFNet [27], FPNet [6], FSPNet [19] and BBRF [28].

Quantitative comparisons. Table 1 reports quantitative comparison results for SFPSeg and 18 SOTA methods on four scratch defect datasets (NEUSD, Severstal, Car, and Chip) in terms of P, R, F1, and mIoU. Our method achieves the best performance on all datasets, establishing new SOTA results with mIoU scores of **91.16%** on NEUSD, **81.40%** on Severstal, **91.01%** on Car, and **78.33%** on Chip. Specifically, our method significantly outperforms existing SDS models, such as SRPCNet [3] and GCRANet [44]. Taking SRPCNet [3] as an example, our method achieves average improvements across all four datasets of +18.41 percentage points (%) in P, +8.95% in R, +13.82% in F1, and +9.26% in mIoU. Notably, on the Severstal dataset, our method outperforms the second-best method, BBRF [28], by +4.16% in P, +0.76% in R, +2.52% in F1, and +1.57% in mIoU. On the Chip dataset, SFPSeg surpasses IdeNet [13] by +10.69% in P and +1.75% in mIoU, with a superior F1 score (+2.28%) albeit at a

Table 1: Comparison with 15 SOTA methods on the NEUSDS, MSD, and Car datasets. The optimal and suboptimal results are highlighted in Red and Blue, respectively.

Methods	Year	NEUSD				Severstal			
		P↑	R↑	F1↑	mIoU↑	P↑	R↑	F1↑	mIoU↑
BBRF [28]	TIP 23	87.68	90.16	88.90	89.40	74.98	79.39	77.12	79.83
FSPNet [19]	CVPR 23	84.64	85.22	84.93	86.02	66.54	50.21	57.23	67.76
FPNet [6]	ACM MM 23	84.49	92.91	88.50	88.81	65.91	74.96	70.14	74.37
MSCAFNet [27]	TCSVT 23	89.53	88.67	89.10	89.73	73.49	77.56	75.47	78.69
IdeNet [13]	TIP 24	87.67	88.40	88.03	88.57	58.22	81.12	67.79	71.67
ZoomNext [30]	TPAMI 24	90.60	88.06	89.31	89.63	75.58	63.36	68.93	74.03
FSEL [37]	ECCV 24	87.27	91.78	89.47	89.77	74.92	76.90	75.90	78.81
CamoDiffusion [4]	AAAI 24	86.59	92.80	89.59	89.91	67.35	81.30	73.67	76.91
PEM [2]	CVPR 24	85.32	84.89	85.10	86.11	74.60	81.47	77.88	79.61
DBCNet [49]	EAAI 25	76.59	85.45	80.78	82.49	57.43	77.53	65.98	71.31
WPFormer [43]	CVPR 25	89.93	90.23	90.08	90.38	75.23	76.68	75.95	78.95
SRPCNet [3]	TII 25	81.64	90.63	85.90	87.03	59.61	70.49	64.60	71.30
SDCT [8]	TIM 25	88.05	93.36	90.63	90.71	77.31	76.59	76.95	79.79
DKD [46]	TII 25	84.39	87.49	85.91	85.21	76.78	70.67	3.60	75.28
RodeconNet [42]	ACM MM 25	84.00	82.07	83.02	84.47	63.95	66.60	65.25	71.77
TextSAM-EUS [36]	ICCV 25	88.16	91.92	90.00	90.23	74.21	75.26	74.73	77.37
DCON [39]	PR 26	86.91	89.76	88.31	88.66	77.72	54.48	64.06	71.62
GCRANet [44]	PR 26	87.39	90.58	88.96	89.44	71.54	71.04	71.29	75.80
Ours	-	90.67	91.41	91.04	91.16	79.14	80.14	79.64	81.40
Methods	Year	Car				Chip			
		P↑	R↑	F1↑	mIoU↑	P↑	R↑	F1↑	mIoU↑
BBRF [28]	TIP 23	76.60	81.55	79.00	82.57	72.90	67.43	70.06	76.26
FSPNet [19]	CVPR 23	63.65	54.76	58.87	71.53	46.86	23.87	31.63	57.66
FPNet [6]	ACM MM 23	52.78	66.08	58.69	69.88	59.62	60.80	60.20	69.34
MSCAFNet [27]	TCSVT 23	79.02	81.67	80.32	84.03	69.22	68.15	68.68	74.58
IdeNet [13]	TIP 24	69.07	84.67	76.08	80.19	69.16	75.51	72.20	76.58
ZoomNext [30]	TPAMI 24	47.70	43.09	45.28	66.04	74.34	52.51	61.55	70.24
FSEL [37]	ECCV 24	77.65	75.46	76.54	81.38	71.17	66.20	68.60	74.52
CamoDiffusion [4]	AAAI 24	85.19	91.18	88.08	89.83	65.68	73.32	69.29	75.01
PEM [2]	CVPR 24	68.28	54.15	60.40	69.97	69.68	57.19	62.82	70.58
DBCNet [49]	EAAI 25	80.02	71.63	75.59	80.70	56.20	57.15	56.67	67.03
WPFormer [43]	CVPR 25	83.74	87.74	85.69	87.88	72.84	69.54	71.15	76.39
SRPCNet [3]	TII 25	69.88	77.05	73.29	79.13	54.11	56.94	55.49	67.40
SDCT [8]	TIM 25	87.48	86.65	87.06	88.01	75.75	57.67	65.48	73.40
DKD [46]	TII 25	57.41	60.53	58.93	64.01	72.84	51.84	60.56	67.27
RodeconNet [42]	ACM MM 25	64.69	74.67	69.32	76.99	50.52	50.70	50.61	64.77
TextSAM-EUS [36]	ICCV 25	65.39	71.99	68.53	75.03	76.51	61.91	68.44	74.32
DCON [39]	PR 26	77.84	70.14	73.79	78.21	68.56	29.67	41.42	61.77
GCRANet [44]	PR 26	78.28	76.60	77.43	81.92	66.94	57.75	62.01	70.93
Ours	-	89.23	89.56	89.39	91.01	79.85	69.78	74.48	78.33

trade-off of −5.73% in recall. These results show that our method maintains superior effectiveness and robustness even on large-scale datasets and those with complex background scenes. Moreover, compared to spatial-frequency dual-domain segmentation methods such as FSEL [37], FPNet [6], and WPFormer [43], our method also exhibits superior performance. For instance, averaged across the four datasets, our method surpasses WPFormer [43] by +4.29% in P, +1.68% in R, +2.92% in F1, and +2.08% in mIoU. Furthermore, our method outperforms TextSAM-EUS [36], a SOTA prompt-based segmentation method that leverages both text and geometric prompts as guidance signals, with average performance gains of +8.66% in P, +7.45% in R, +8.22% in F1, and +6.24% in mIoU across the four datasets.

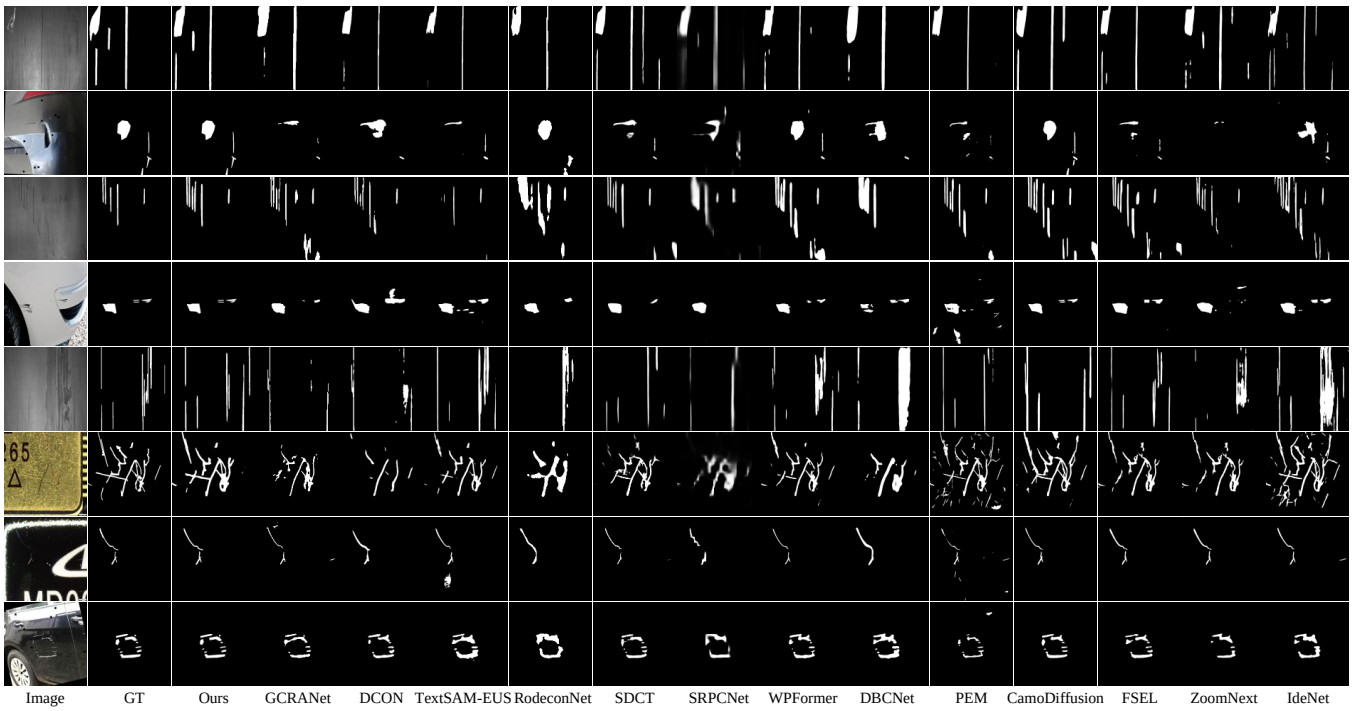


Figure 3: Visual comparison of segmentation results on the SDS datasets.

Visualization comparisons. Figure 3 presents a visual comparison of segmentation results between our method and existing representative approaches across the four SDS datasets. First (rows 1-2), some methods fail to detect complete scratches, particularly those that are faint or exhibit low contrast against the background. Second (rows 3-4), they are prone to misclassifying background textures as scratches due to their visual similarity. Third (rows 5-6), for scratches with abrupt morphological variations or physical discontinuities, existing methods often erroneously generate continuous segmentation masks. Finally (last two rows), they struggle to capture fine-grained scratch details. In contrast, our method provides more accurate and robust predictions, excelling in segmenting scratches with varying morphologies, effectively suppressing background interference, and preserving richer scratch details.

4.4 Ablation and Analysis

Components Incremental Ablations.

The ablation studies of the proposed components on the Car (smallest) and Severstal (largest, with complex scenes) are summarized in Table 2. The baseline model, comprising the backbone, a basic decoder (with two $\mathrm{Up}_{2\times}$ layers, each followed by a $\mathrm{BottConv}_3$ block), and a segmentation head, achieves mIoU scores of 86.89% and 80.57% on Car and Severstal, respectively. First, incorporating the proposed BFF module improves the mIoU by +0.20% on Car and +0.40% on Severstal, with only a marginal increase of +0.01M parameters and +0.04 GFLOPs. Subsequently, replacing the basic decoder with our MPLD brings further gains: the mIoU increases by +4.04% on Car and +0.25% on Severstal, while the recall improves by +5.08% and +1.73%, respectively. Overall, the complete SFPSeg model achieves mIoU scores of 91.01% on the Car dataset and 81.40% on the Severstal dataset, reaching SOTA performance.

Spatial/Frequency Branches within MPLD. We conduct three ablation experiments to verify the impact of the spatial and frequency branches in the MPLD, as detailed in Table 3. The Spatial branch uses only the MSP module without the parallel frequency branch. The Frequency branch uses only the WPF module without the parallel spatial branch, resulting in a purely frequency-domain decoder. The Spatial+Frequency combines both branches, representing the proposed MPLD in SFPSeg. Results show that the Spatial branch, which introduces a Mamba structure to capture global features, incurs higher parameter counts and FLOPs but yields a substantial performance gain (+2.18% mIoU on Car). The Frequency branch has the lowest parameters and FLOPs while achieving significant scratch defect segmentation metrics. The combined Spatial+Frequency method achieves the best performance. It outperforms the individual branches by +3.92% mIoU on the Car dataset and by +0.43% mIoU on the Severstal dataset.

Interaction mechanisms of the MSP and the WPF. Table 4 (a) shows the effect of different interaction mechanisms of the MSP and the WPF within MPLD, i.e., additive fusion and concatenate fusion mechanism adopted in our MPLD. Compared with additive fusion, concatenation yields superior performance (e.g., +2.95% on Car in mIoU). This advantage may be attributed to its ability to preserve the specificity of heterogeneous features (spatial-domain features from MSP and frequency-domain features from WPF) while enabling adaptive cross-modal fusion in subsequent layers.

Global/Local Branches within MSP. Table 4 (b) shows the impact of the local and global branches in MSP. The results indicate

Table 2: Incremental Ablations

Baseline	BFF	MPLD	Params(M)	FLOPs(G)	Car				Severstal			
					P↑	R↑	F1↑	mIoU↑	P↑	R↑	F1↑	mIoU↑
✓			25.47	101.06	85.92	83.69	84.79	86.89	78.35	78.76	78.55	80.57
✓	✓		+0.01	+0.04	82.46	86.77	84.56	87.09	78.45	79.79	79.11	80.97
✓		✓	+1.61	+12.76	88.89	88.77	88.83	90.79	77.69	**80.49**	79.07	80.84
✓	✓	✓	+1.61	+12.78	**89.23**	**89.56**	**89.39**	**91.01**	**79.14**	80.14	**79.64**	**81.40**

Table 3: Ablation study within MPLD

Method	Params(M)	FLOPs(G)	Car				Severstal			
			P↑	R↑	F1↑	mIoU↑	P↑	R↑	F1↑	mIoU↑
Spatial (MSP)	+1.21	+10.86	87.35	88.67	88.01	89.27	78.26	**80.96**	79.59	81.22
Frequency (WPF)	+0.4	+2.26	89.11	89.47	89.29	90.73	**80.06**	78.13	79.08	80.91
Spatial+Frequency (MSP+WPF)	+1.61	+12.78	**89.23**	**89.56**	**89.39**	**91.01**	79.14	80.14	**79.64**	**81.40**

Table 4: Ablation studies of key components in SFPSeg. (a) Effects of interaction mechanisms within MPLD; (b) Effects of global and local branches in the MSP module; (c) Effects of fusion strategies across frequency components in the WPF.

	Settings	Car				Severstal			
(a)	MPLD	P↑	R↑	F1↑	mIoU↑	P↑	R↑	F1↑	mIoU↑
	w/ Add	88.55	84.60	86.53	88.56	77.84	**81.08**	79.43	81.09
	w/ Concatenate	**89.23**	**89.56**	**89.39**	**91.01**	**79.14**	80.14	**79.64**	**81.40**
(b)	MSP	P↑	R↑	F1↑	mIoU↑	P↑	R↑	F1↑	mIoU↑
	w/ Local	88.29	88.15	88.22	90.16	79.56	79.61	79.58	81.26
	w/ Global	88.08	89.53	88.80	90.73	**79.90**	79.28	79.59	81.28
	w/ Local + Global	**89.23**	**89.56**	**89.39**	**91.01**	79.14	**80.14**	**79.64**	**81.40**
(c)	WPF	P↑	R↑	F1↑	mIoU↑	P↑	R↑	F1↑	mIoU↑
	w/ Concatenate	74.21	89.46	81.12	84.40	79.39	79.14	79.26	81.11
	w/ Add	**89.92**	86.78	88.32	90.48	**79.72**	79.14	79.43	81.25
	w/ Concatenate + Add	89.23	**89.56**	**89.39**	**91.01**	79.14	**80.14**	**79.64**	**81.40**

that by fusing the complementary features from both branches, the model achieves the best overall performance, effectively balancing detail sensitivity and holistic scene understanding.

Frequency Fusion Strategies within WPF. Table 4 (c) presents the results of frequency fusion strategies in WPF. While additive fusion leads in Precision, our adaptive fusion strategy achieves the best over performance, excelling in both F1-score and mIoU. For example, on the Car dataset, it outperforms simple concatenation with gains of +6.60% in P, +0.10% in R, +8.27% in F1, and +6.61% in mIoU. These improvements are attributed to our spatially adaptive design, which dynamically fuses frequency subbands via learned inter-band communication.

5 Conclusion

This paper proposes SFPSeg, a novel framework that unifies prompt learning in both spatial and frequency domains for scratch defect segmentation. The core of the SFPSeg is a Mixed-domain Prompt Learning Decoder (MPLD), which decodes complementary scratch patterns by integrating two novel components: a Morphology-aware Spatial Prompting module that precisely captures structural details, and a Wavelet Prompt Filter that amplifies defect-related frequency signals while suppressing background responses in the wavelet domain. Extensive experiments on multiple benchmarks demonstrate the state-of-the-art performance of SFPSeg.

Limitations. As an initial exploration of prompt learning in the spatial-frequency dual domain for defect segmentation, this work focuses on a learnable continuous prompt. The potential of semantic prompts (e.g., text-based guidance) has not yet been explored, presenting a clear direction for future work.

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